

Phone-level accentedness scoring

From speech and expected phonemes to an interpretable

0-100 score for every sound.

AUDIO + PHONES

ORDINAL MODEL

GRADIO DEMO

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Modeling • evaluation • product demo

ILLUSTRATIVE OUTPUT



hello

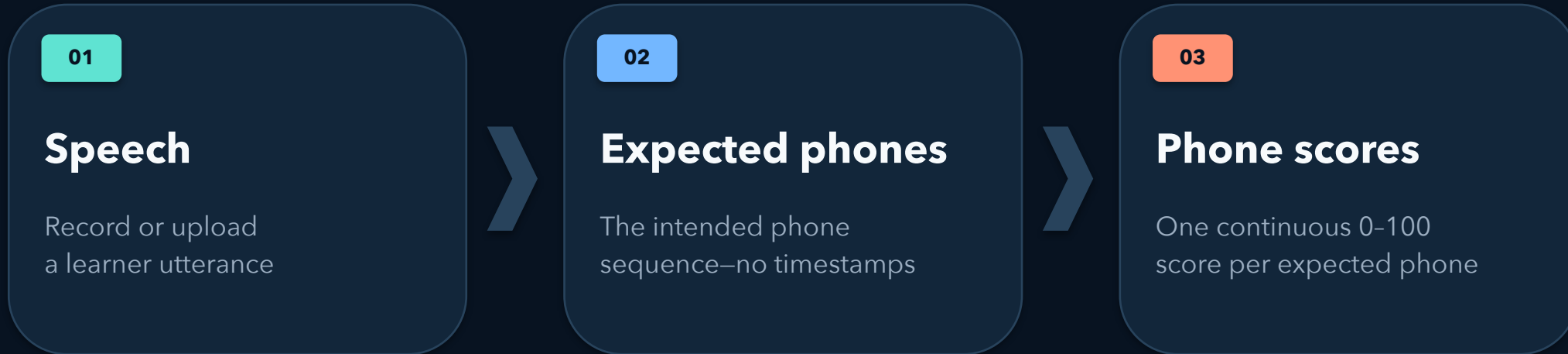
expected phones

h	ə	l	ou
92	45	89	12

$$\text{Score} = 50 \times [P(Y \geq 1) + P(Y \geq 2)]$$

Source: challenge brief example; scores shown are illustrative.

The task: localize pronunciation feedback



What makes it difficult

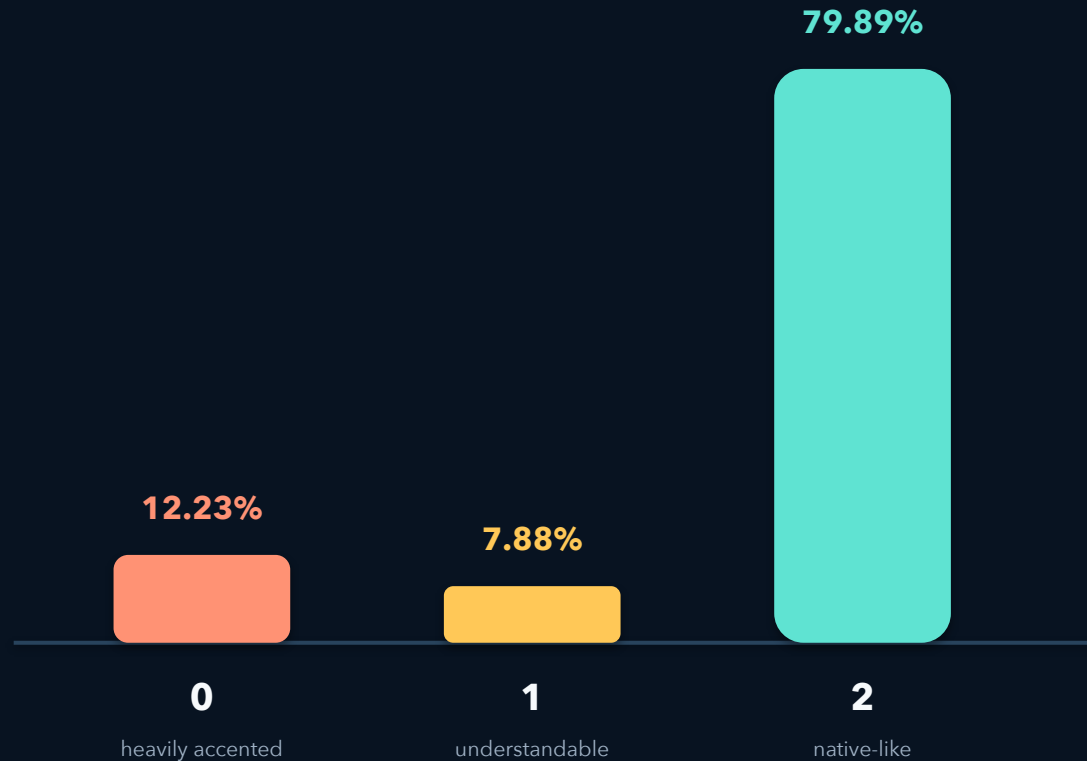
- Labels are ordered ($0 < 1 < 2$), but output must be continuous.
- Phone boundaries are absent and must be inferred from audio.
- The majority label dominates, so raw accuracy is misleading.

Product question

Where did this realization sound less American-like—and what should the learner practice next?

The data is small—and heavily imbalanced

TRAIN LABEL DISTRIBUTION



NO PHONE TIMESTAMPS

NO SPEAKER IDs

NO RATER AGREEMENT

2,799

TRAIN UTTERANCES

100

VALIDATION UTTERANCES

90,239

LABELED PHONES TOTAL

44

PHONE TYPES

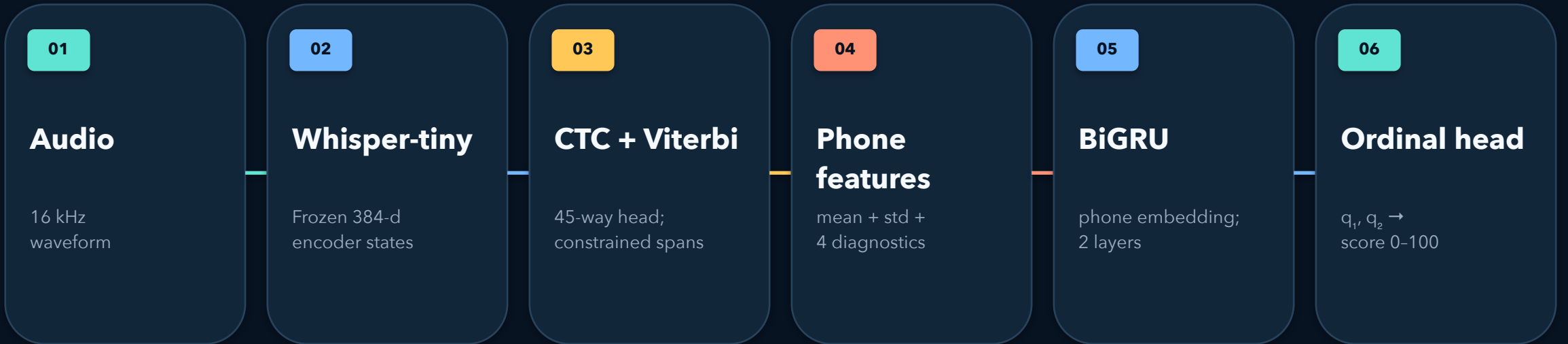
79.47%

validation accuracy by always predicting
"native-like"

Yet recall for labels 0 and 1 is zero.

These gaps shape both modeling and evaluation.

Architecture: align first, then score

**9**

CTC epochs

18

scorer epochs

0

alignment fallbacks

42

random seed

Training discipline: tune epochs on train-only development data, then restart and fit all 2,799 training utterances.

The objective and metrics match the ordering

ORDINAL REGRESSION

Two ordered questions

$$q_1 = P(Y \geq 1) \quad q_2 = P(Y \geq 2)$$



$$\text{score} = 50 \times (q_1 + q_2)$$

EVALUATION

Balanced MAE

Primary metric

Compute MAE within each true label, then average the three class errors equally.

- Preserves continuous error magnitude
- Stops the 80% majority class dominating

Secondary checks

QWK

MACRO-F1

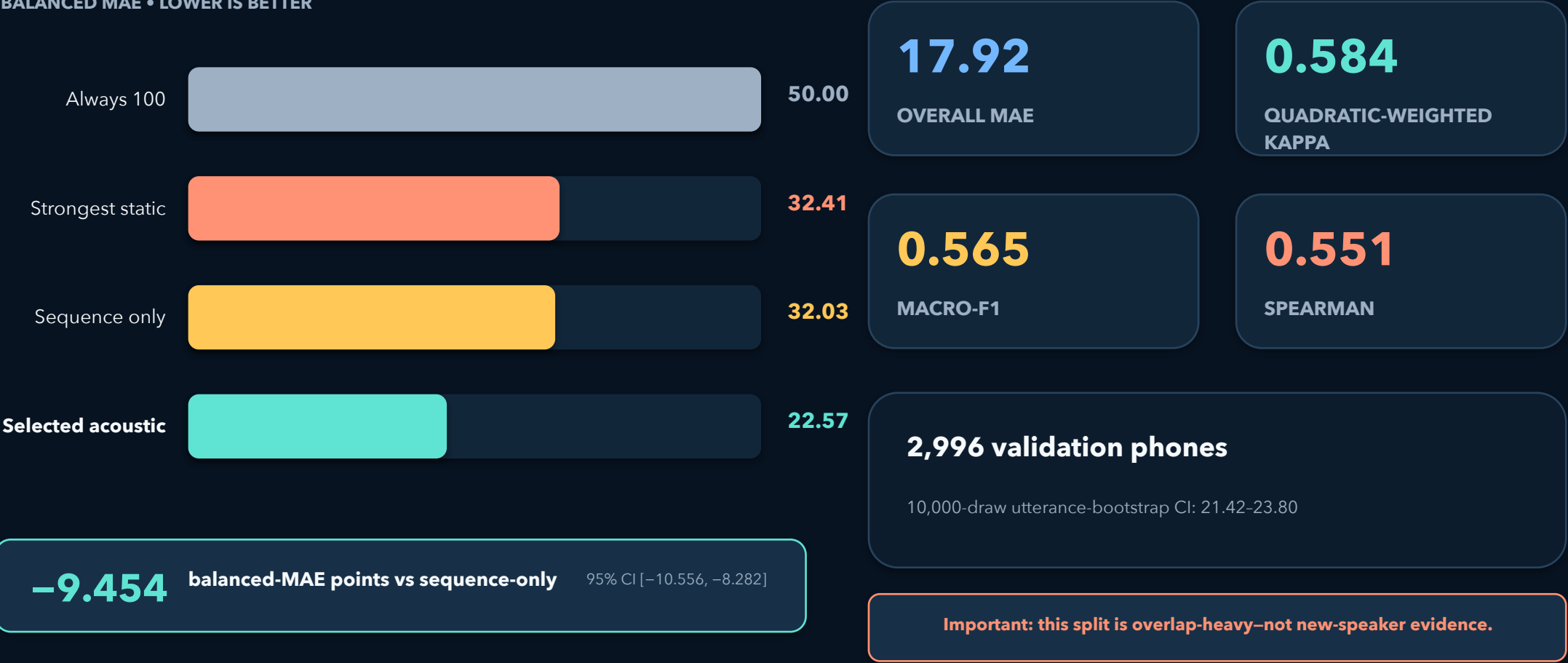
BALANCED ACC.

SPEARMAN

Loss weighting: inverse $\sqrt{\text{class frequency per phone token}}$

Validation: audio adds material signal

BALANCED MAE • LOWER IS BETTER



What I tried—and deliberately did not ship

REJECTED

Bigger encoder

25.60 vs 22.57

Whisper-small worsened balanced MAE; no small-specific hyperparameter retuning was run.

REJECTED

Auxiliary labels

+0.039 BM delta

Severity and cluster targets produced no reliable gain; class-0 MAE worsened.

REJECTED

Full inverse weights

BM -2.02 • MAE +3.86

Rare-class recall improved, but QWK, macro-F1, majority recall, and calibration regressed.

SELECTED

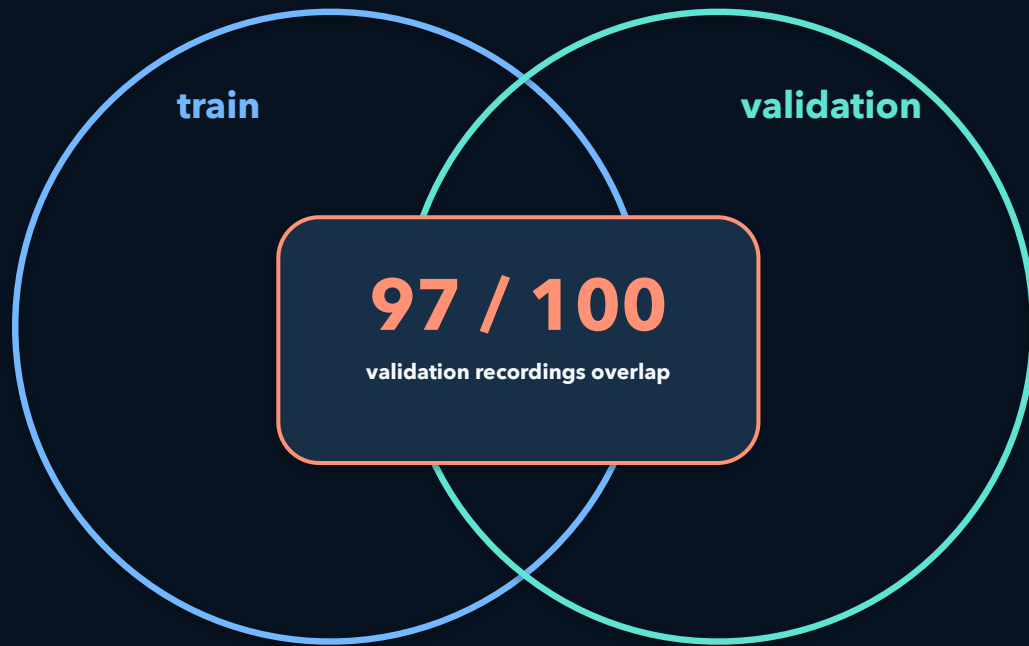
Frozen Whisper-tiny + constrained CTC + inverse- $\sqrt{\cdot}$ weighted ordinal head

The simplest controlled candidate had the best overall trade-off.

Each result comes from its documented protocol; this is not a single shared leaderboard.

The largest risk is the validation split

PSEUDO-SPEAKER OVERLAP



98.0% of validation phones

sit inside those inferred voice clusters

What this changes

- 92 / 100 validation prompts also appear in training.
- Reported metrics describe a partly seen-speaker / seen-prompt distribution.
- A speaker-disjoint benchmark is required before making generalization claims.

WavLM clusters are inferred voices—not verified identities.

Sniff test: right direction, weak separation

CONTROLLED OWN-VOICE PAIR

+3.13
mean points

Best American



70.05

Non-native



66.92

- Only 10 / 20 phones moved higher
- Pace differed: 2.76s vs 4.08s

HEAVILY ACCENTED PHONES

43.53%

CORRECTLY SCORED
BELOW 25

11.94%

INCORRECTLY SCORED $\geq$
75

Failure pattern

The model reacts when accent also hurts expected-phone recognition, but can miss subtler non-American realizations that Whisper still recognizes confidently.

Conclusion: marginal utterance-level pass—not a phone-level pass.

Automated judges did not validate the labels

GOPT teacher

5,863 / 5,894 → label 2

Useful ranking signal, but cross-corpus calibration collapsed near native-like; macro-F1 0.299.

NOT SAFE TO RELABEL

Local audio LLM

347 / 347 → label 2

Gemma 4 12B passed structure checks but failed the informativeness gate completely.

NOT SAFE TO RELABEL

GPT audio judge

40% agreement • 0 label-0

Balanced 30-item audit: macro-F1 0.299, QWK 0.143; it remained too lenient.

NOT SAFE TO RELABEL

Next valid step

Blinded expert review. The deterministic packets exist; no human-rating ledger exists yet.

Demo: turn a score into a practice loop

Pronunciation practice

Sentence to practice

We are both children together.

Generated phones • editable

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Record / upload audio

Score pronunciation

Score bands

practice

developing

strong

Product choices

- Microphone or upload
- Sentence playback
- Automatic, editable phonemes
- Ordered per-phone feedback

Difficulty ≠ model score

Keep raw scores fixed; adapt only the coaching threshold per phone and learner.

Run locally: `cd submission • uv run python demo_app.py`

Takeaway: useful signal, honest uncertainty

The model learned **useful acoustic signal**.

The evaluation has not yet proven **new-speaker robustness**.

01

Benchmark

Collect a genuinely speaker-disjoint, expert-rated evaluation set.

02

Calibrate

Review label disagreements and fit phone-specific calibration with uncertainty.

03

Improve

Test a more accent-sensitive encoder across seeds and matched-pace voice pairs.

Ship today

A coaching prototype with visible uncertainty—not a high-stakes judgment of identity, ability, or proficiency.

[Questions](#) • [discussion](#) • [live demo](#)